

Desert Ecosystems

Starter

An ecosystem is a network of interactions between all the living organisms in a habitat and the non-living things around them

1. Define an ecosystem?

2. In which direction do the arrows go in a food chain?

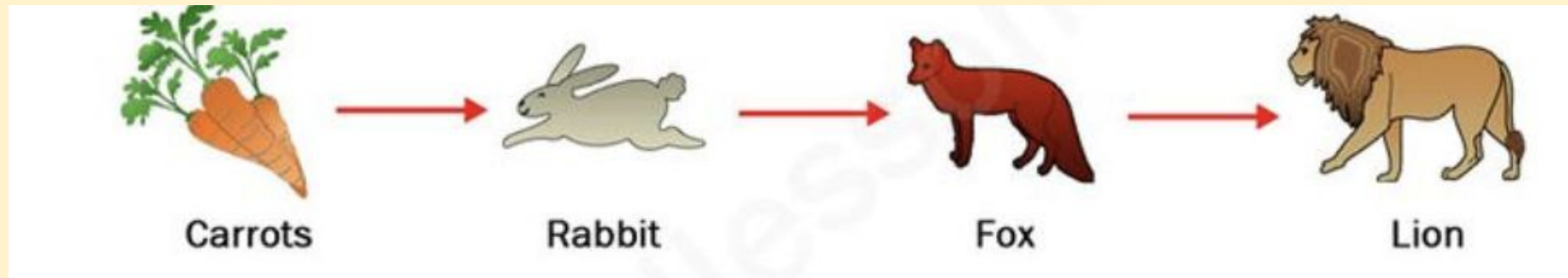


3. Give an example of an ecosystem and a food chain you might find there

Desert Ecosystems

Starter

1. Define an ecosystem?
An ecosystem is a network of interactions between all the living organisms in a habitat and the non-living things around them
2. In which direction do the arrows go in a food chain?



3. Give an example of an ecosystem and a food chain you might find there

Success criteria

Lesson Title	Working towards	Working at	Working beyond
What is an Ecosystem	I can name some living (biotic) and non-living (abiotic) parts of an ecosystem.	I can define key terms like community, population, habitat and explain how energy moves through a food web	I can explain the role of decomposers and interpret pyramids of numbers to show energy transfer in ecosystems.
Interactions within Ecosystems	I can identify some habitats found in the Sonoran Desert and other ecosystems.	I can describe interactions between organisms and their environment in different ecosystems.	I can explain the differences between habitats and ecosystems and how organisms are adapted to survive in these settings.
Adaptations and Competition	I can give examples of how plants or animals are adapted to their environments.	I can describe how adaptations help species survive and explain competition between organisms.	I can analyse mutualistic relationships and present research on how different species interact in ecosystems.
Invasive Species	I can name an invasive species and describe what a native species is.	I can describe how invasive species affect native plants and animals in an ecosystem.	I can explain why controlling invasive species is important and evaluate different methods of control.
Case Study - New Zealand (Debate)	I can identify an invasive species problem in New Zealand.	I can describe the ecological impact of invasive species and outline arguments for control.	I can construct and defend an argument in a debate using evidence about species control and conservation.
Bioaccumulation	I can state what an insecticide is and give a simple example.	I can describe how chemicals like DDT build up in organisms and impact the food chain.	I can explain why organisms at the top of the food chain have higher toxin levels and the environmental consequences.
Weather and Climate	I can describe the difference between weather and climate.	I can explain how weather affects plants and animals in an ecosystem.	I can analyse weather data to predict effects on local ecosystems and seasonal changes.
Climate and Atmosphere	I can name key gases in the atmosphere.	I can explain how greenhouse gases affect Earth's climate and describe human impacts.	I can evaluate how climate change may affect ecosystems and propose possible solutions.

Desert Ecosystems

What is special about a desert ecosystem?



Comprehension Task

Read the text about animal survival in deserts and answer the questions

We will discuss the answers after



Desert Animal Survival

Read the information and then answer the questions.

Lack of water creates a survival problem for all desert organisms - animals and plants alike. Animals have a bigger problem than plants. They are more susceptible to extremes of temperature than plants.



Animals receive heat from the sun, from rocks and soil and also from the hot air in the desert. An animal's body can only survive in a narrow range of temperatures. If it gets too hot or too cold the animal will die.

For many months in the desert the temperature is very, very hot during the day and then also very cold at night.

Most desert animals have evolved ways of dealing with the heat problems and the lack of water in the desert. Each different animal does this in a different way so there are many thousands of different ways that animals deal with heat and lack of water.

Some animals conserve water and many others actually make water.

1. What 2 main survival problems do desert animals have?

2. Why do animals struggle in the desert more than plants?

3. What will cause an animal to die in the desert?

4. How do animals survive in the desert?

How many deserts can you name?



(not desserts!)





What do these deserts have in common?

What differences do they have?

FACTS

- The largest hot desert in the world – Sahara (northern Africa), reaches temperatures of up to 50°C during the day.
- The Gobi desert in Asia and the polar deserts of the Antarctic and Arctic, which are the world's largest, are always cold.
- Chile's Atacama desert is the driest in the world (after Antarctica) having parts that receive < 2 mm of precipitation/year.
- Namib Desert (Southern Africa) is believed to be as old as 55 million years; the oldest desert in the world.
- Thar Desert (India) is believed to be the most civilised desert in the world due to having roads, infrastructure and human settlements dating back to thousands of years (about 50,000 years old).

The Sonoran Desert

Saguaro cacti grow very slowly and can be over 100 years old.

How are they adapted to their environment?

- Their roots spread out widely just underneath the soil
- They have no leaves
- Large stem to store water
- Thick waxy skin



Interactions in the Sonoran Desert

Saguaro cacti can create habitats for other animals

- Gila Woodpeckers make holes in the cacti to make nests
- Other birds then also use these holes

Cactus wrens like to use a different type of cactus called a teddy bear cholla to make their nests. They are very spiky so not many other animals will go near them. Why would this be beneficial to cactus wrens?

It means their eggs are protected from predators!!



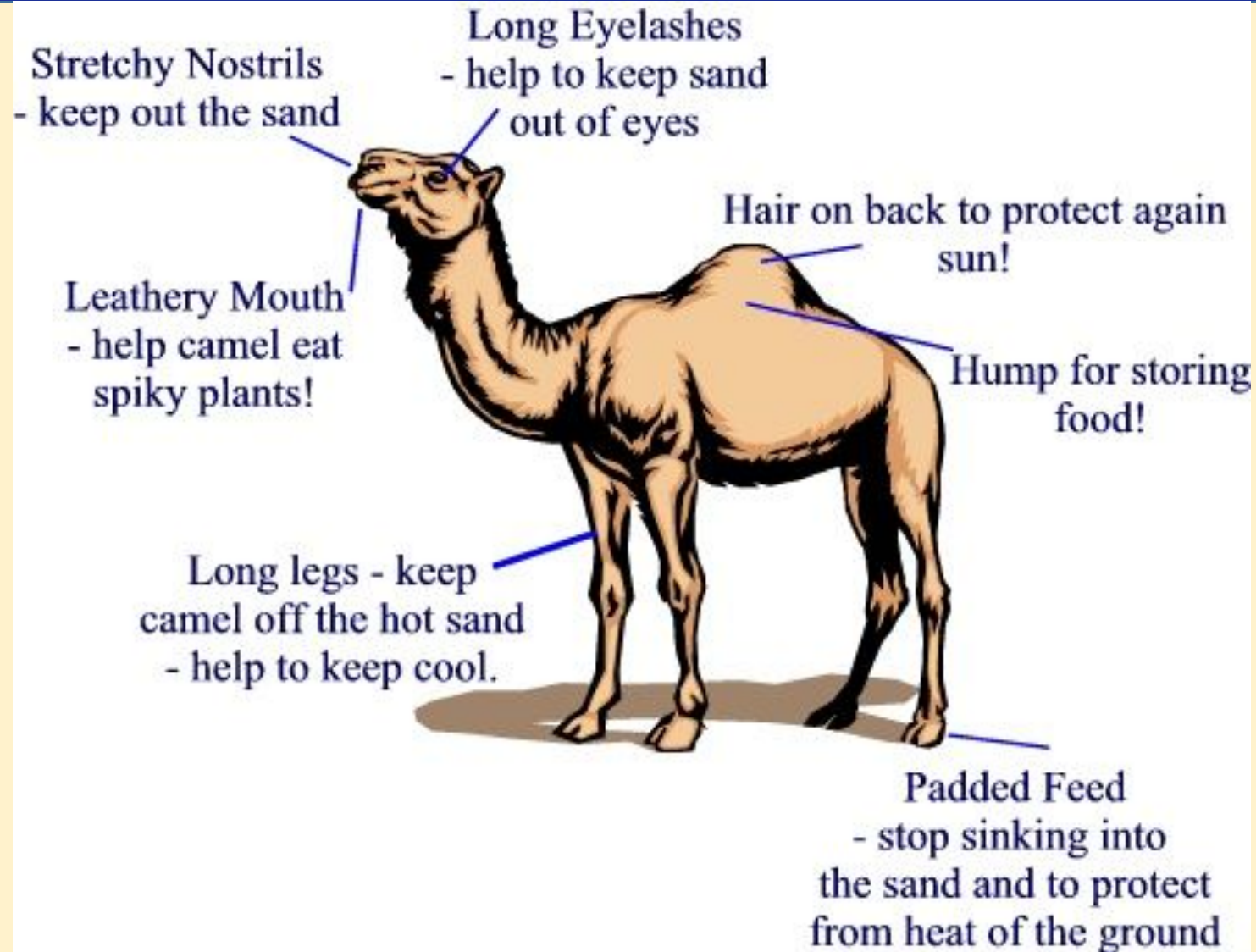
Interactions in the Sonoran Desert



Camels are also found in the Sonoran Desert.

How are they adapted to their environment?

Interactions in the Sonoran Desert

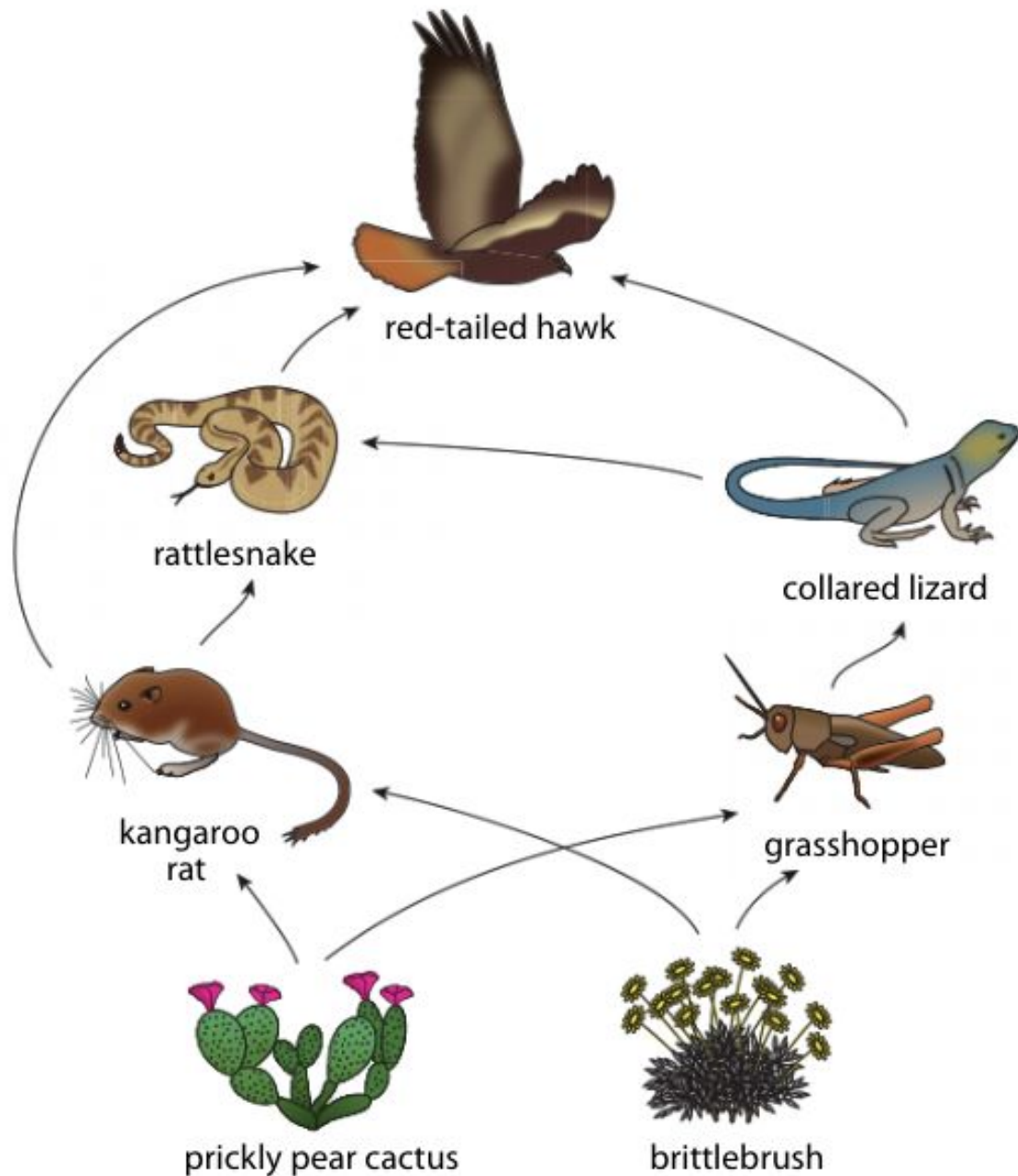


Non-living factors

Discuss with your partner, how would these factors affect organisms in the desert?

- Light **Needed for photosynthesis**
- Temperature **It is very hot! Why are many animal nocturnal or bury holes in the ground/ sand?**
- Soil **Minerals for plants, building materials for animals**
- Water **All organisms need water! It might only rain once a year**
- Air **Needed for photosynthesis and respiration**

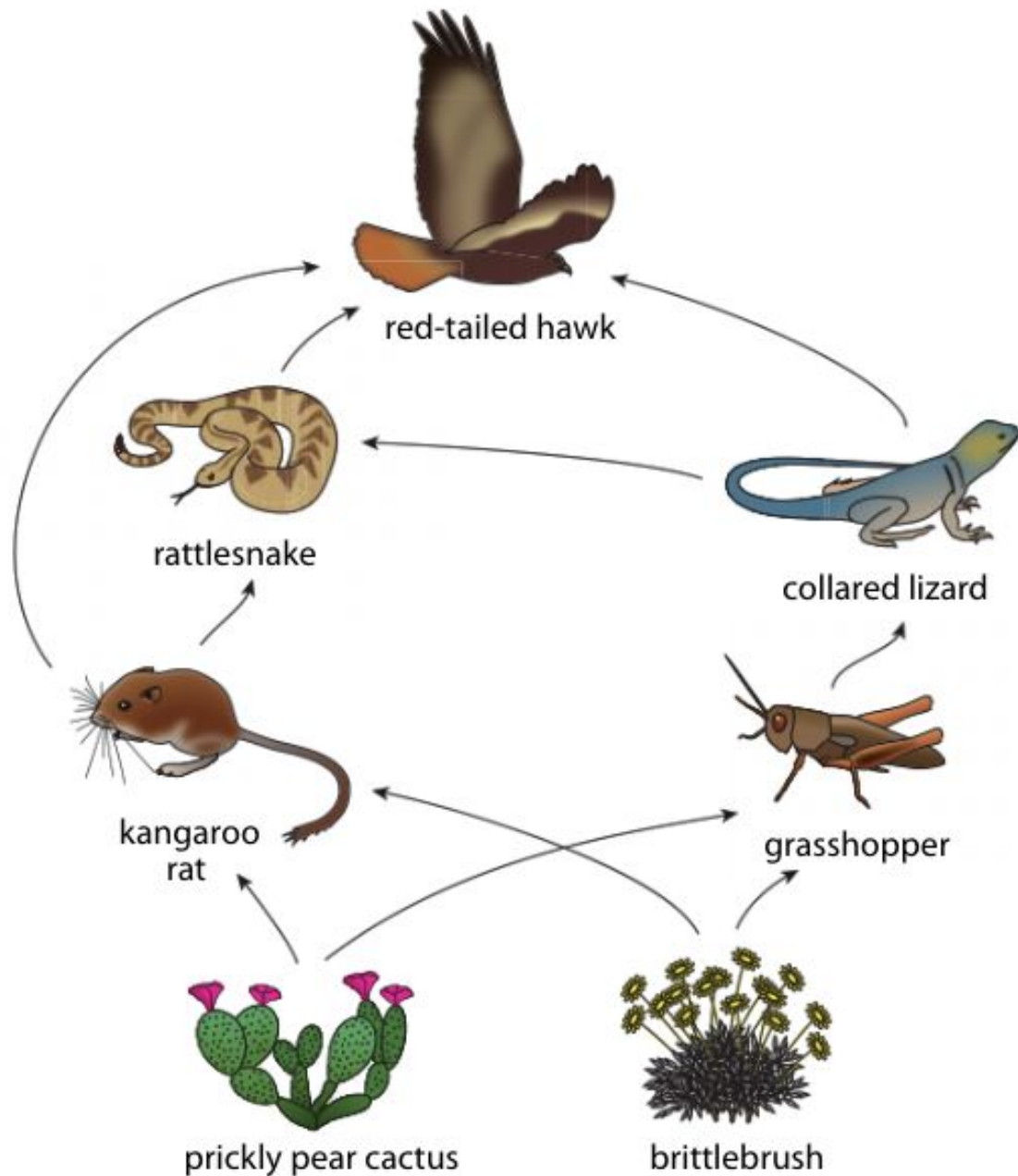




How many food chains can you see?

How many producers are there?

Can you explain the grasshoppers role in this food web?



How many food chains can you see?

6

How many producers are there?

2

Can you explain the grasshoppers role in this food web?

The grasshopper is a **primary consumer** (herbivore). It **eats producers** (like the brittlebrush) to gain energy and is then **eaten by secondary consumers** such as the **collared lizard**.

Task: Create a 'wanted' poster

Research one desert animal to create a wanted poster for

Include:

- What it eats and what eats it
- The habitat it lives in
- How the species is adapted to survive in its habitat
- How its actions affect other organisms or non-living parts of the ecosystem (not just feeding)
- How other organisms or non-living things affect it



Wanted: Fenec Fox! 
Name: Fenec Fox Description of Animal: Fenec Foxes are the smallest of all foxes. They are 14-17 inches long when full grown. They weigh between 2 and 3.5 pounds. They have huge ears and large eyes. Their fur is sandy-colored. Where Does it Live? The Sahara Desert, and other sandy deserts. What Does it Eat? They are omnivores. They eat plants and animals. Some of the things they eat are: berries, fruit, small mammals, birds, eggs and lizards. Adaptations: 1. Big Ears: Helps his body cool off. 2. Fur on the bottom of his feet: Helps keep his feet from burning on the desert sand. 3. Nocturnal: He comes out at night when it is cooler. 4. Color of Fur: Blends in with desert sand 5. Large eyes: Helps him see well at night.

Task: Evaluation

Questions:

1. Where did you find the best information?
2. Which websites were most relevant for your research?
3. How did you choose which websites to use?

